

Shift-invariant functions that behave like polynomials

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Definition

$S : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is the cyclic left-shift:

$$S(x_1, \dots, x_n) = (x_2, \dots, x_n, x_1)$$

A function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is called shift-invariant if for all $x \in \mathbb{F}_2^n$ it holds that

$$f(S(x)) = S(f(x)).$$

Shift-invariant permutations are not well-understood:

- Many proofs are ad-hoc
- Very few general constructions are known

The map χ

Definition

The map $\chi : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n, x \mapsto y = \chi(x)$ is given by

$$y_i = x_i + x_{i+2}(1 + x_{i+1}),$$

where the indices are taken modulo n .

The map χ

Definition

χ is given by

$$y_i = x_i + x_{i+2}(1 + x_{i+1}).$$

Observation

It holds that $y_i \in \{x_i, x_i + 1\}$ and

$$y_i = x_i + 1 \quad \Longleftrightarrow \quad (x_{i+1}, x_{i+2}) = (0, 1).$$

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Definition

We say that χ is given by the complementing landscape $*01$.

The map χ

Example

$n = 9$

$x = 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0$

$\chi(x) =$

The map χ

Example

$n = 9$

$$\begin{array}{rcccccccccc} x & = & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ & & \downarrow & & & & & & & & \\ \chi(x) & = & 1 & & & & & & & & \end{array}$$

The map χ

Example

$n = 9$

$$\begin{array}{rcccccccccc} x & = & 1 & \textcolor{red}{1} & \textcolor{green}{0} & \textcolor{green}{1} & 1 & 0 & 1 & 0 & 0 \\ & & & \downarrow & & & & & & & \\ \chi(x) & = & 1 & 0 & & & & & & & \end{array}$$

The map χ

Example

$n = 9$

$$\begin{array}{rcccccccccc} x & = & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ \chi(x) & = & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & \end{array}$$

↓

The map χ

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$n = 9$

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The map χ

- First introduced by Joan Daemen¹.
- χ is a permutation on $\mathbb{F}_2^n$ if and only if n is odd¹.
- Used in several cryptographic primitives, for example
 - SHA-3 (Keccak)
 - ASCON
 - Subterranean
 - Rasta

¹Joan Daemen. “Cipher and hash function design strategies based on linear and differential cryptanalysis”. PhD thesis. Doctoral Dissertation, March 1995, KU Leuven, 1995.

Iterates of χ

We can compute the iterates of χ , i.e. what is

$$\chi^k(x) = \chi(\chi(\dots \chi(x) \dots))$$

for $k \geq 1$.

Fact

We have

$$x_i \cdot (1 + x_i) = 0$$

because we are working in $\mathbb{F}_2$.

The first few iterates:

$$\chi^1(x)_i = x_i + x_{i+2}(1 + x_{i+1}),$$

Iterates of χ

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$$\chi^1(x)_i = x_i + x_{i+2}(1 + x_{i+1}),$$

$$\chi^2(x)_i = x_i + x_{i+4}(1 + x_{i+3})(1 + x_{i+1}),$$

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$$\chi^2(x)_i = x_i + x_{i+4}(1 + x_{i+3})(1 + x_{i+1}),$$

$$\begin{aligned}\chi^3(x)_i = & x_i + x_{i+2} \cdot (1 + x_{i+1}) \\ & + x_{i+4} \cdot (1 + x_{i+3}) \cdot (1 + x_{i+1}) \\ & + x_{i+6} \cdot (1 + x_{i+5}) \cdot (1 + x_{i+3}) \cdot (1 + x_{i+1})\end{aligned}$$

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$$\chi^4(x)_i = x_i + x_{i+8} \cdot (1 + x_{i+7}) \cdot (1 + x_{i+5}) \cdot (1 + x_{i+3}) \cdot (1 + x_{i+1}).$$

We define $\gamma_0 = \text{id}$ and $\gamma_{2k} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ for $k \geq 1$ given by

$$\gamma_{2k}(x)_i = x_{i+2k} \cdot (1 + x_{i+2k-1}) \cdot (1 + x_{i+2k-3}) \cdots (1 + x_{i+1}).$$

Iterates of χ

We define $\gamma_0 = \text{id}$ and $\gamma_{2k} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ for $k \geq 1$ given by

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With that notation we have

$$\chi^0 = \gamma_0$$

$$\chi^1 = \gamma_0 + \gamma_2$$

$$\chi^2 = \gamma_0 + \gamma_4$$

$$\chi^3 = \gamma_0 + \gamma_2 + \gamma_4 + \gamma_6$$

$$\chi^4 = \gamma_0 + \gamma_8.$$

- Study the functions γ_{2i}
- Explain their behavior under composition
- Deduce properties about χ and related functions
- Generalize

Key Lemma

It holds that

$$\gamma_{2k} \circ \left(\gamma_0 + \sum_{i=1}^s a_i \gamma_{2i} \right) = \sum_{i=0}^s a_i \gamma_{2i+2k}.$$

Key Lemma

It holds that

$$\gamma_{2k} \circ \left(\gamma_0 + \sum_{i=1}^s a_i \gamma_{2i} \right) = \sum_{i=0}^s a_i \gamma_{2i+2k}.$$

If γ_0 is not included, then the result does not hold. For example

$$\gamma_2(\gamma_2(x))_i = x_{i+4}(1 + x_{i+3})$$

is not a linear combination of $\gamma_0, \gamma_2, \gamma_4, \dots$

Definition

Let $\mathcal{G}_n$ denote the set

$$\mathcal{G}_n = \gamma_0 + \text{span}\{\gamma_2, \gamma_4, \dots\}.$$

Note that $\chi = \gamma_0 + \gamma_2 \in \mathcal{G}_n$.

Compositions of the functions γ_{2i}

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Note that $\chi = \gamma_0 + \gamma_2 \in \mathcal{G}_n$.

The Key Lemma implies:

Theorem

$\mathcal{G}_n$ is closed under composition. Equivalently, $(\mathcal{G}_n, \circ)$ is a monoid.

Compositions of the functions γ_{2i}

Theorem

$\mathcal{G}_n = \gamma_0 + \text{span}\{\gamma_2, \gamma_4, \dots\}$ is a monoid.

For a monoid $(M, *)$ with identity 1 we can define the unit group

$$M^* = \{m \in M : \exists n \in M : m * n = n * m = 1\}$$

Question

What is the unit group $\mathcal{G}_n^*$? In other words, which maps from $\mathcal{G}_n$ are permutations on $\mathbb{F}_2^n$?

Compositions of the functions γ_{2i}

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Example

$$\gamma_2 \circ (\gamma_0 + \gamma_4) = \gamma_2 + \gamma_6$$

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Example

$$\begin{aligned} \gamma_2 \circ (\gamma_0 + \gamma_4) &= \gamma_2 + \gamma_6 \\ \gamma_4 \circ (\gamma_0 + \gamma_2 + \gamma_6) &= \gamma_4 + \gamma_6 + \gamma_{10}. \end{aligned}$$

Compositions of the functions γ_{2i}

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This looks like polynomial multiplication!

Polynomial interpretation

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Example

$$\gamma_2 \circ (\gamma_0 + \gamma_4) = \gamma_2 + \gamma_6$$

$$X^2 \cdot (1 + X^4) = X^2 + X^6$$

and

$$\gamma_4 \circ (\gamma_0 + \gamma_2 + \gamma_6) = \gamma_4 + \gamma_6 + \gamma_{10}$$

$$X^4 \cdot (1 + X^2 + X^6) = X^4 + X^6 + X^{10}.$$

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and

$$\begin{aligned}\gamma_4 \circ (\gamma_0 + \gamma_2 + \gamma_6) &= \gamma_4 + \gamma_6 + \gamma_{10} \\ X^4 \cdot (1 + X^2 + X^6) &= X^4 + X^6 + X^{10}.\end{aligned}$$

Observation: γ_{2i} seems to behave like X^{2i} .

Polynomial interpretation

We remove the factor of 2 in the exponent of the polynomials.

Example

$$\gamma_2 \circ (\gamma_0 + \gamma_4) = \gamma_2 + \gamma_6$$

$$X \cdot (1 + X^2) = X + X^3$$

and

$$\gamma_4 \circ (\gamma_0 + \gamma_2 + \gamma_6) = \gamma_4 + \gamma_6 + \gamma_{10}$$

$$X^2 \cdot (1 + X + X^3) = X^2 + X^3 + X^5.$$

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Do we get a map

$$\sum_{i=0}^s a_i \gamma_{2i} \mapsto \sum_{i=0}^s a_i X^i?$$

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Fact

If n is odd:

$$\gamma_{n+1} = 0 \iff X^{(n+1)/2} = 0.$$

Polynomial interpretation

Do we get a map

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Almost!

Fact

If n is odd:

$$\gamma_{n+1} = 0 \iff X^{(n+1)/2} = 0.$$

If n is even:

$$\gamma_{2n} + \gamma_n = 0 \iff X^n + X^{n/2} = 0.$$

Polynomial interpretation

We set

$$\mathcal{R}_n = \mathbb{F}_2[X]/(m(X))$$

where

$$m(X) = \begin{cases} X^{(n+1)/2} & n \text{ is odd,} \\ X^n + X^{n/2} & n \text{ is even.} \end{cases}$$

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$$m(X) = \begin{cases} X^{(n+1)/2} & n \text{ is odd,} \\ X^n + X^{n/2} & n \text{ is even.} \end{cases}$$

We have a bijection $\varphi : \text{span}\{\gamma_0, \gamma_2, \gamma_4, \dots\} \rightarrow \mathcal{R}_n$ given by

$$\varphi \left(\sum_{i=0}^s \alpha_i \gamma_{2i} \right) = \sum_{i=0}^s \alpha_i X^i.$$

$\mathcal{G}_n = \gamma_0 + \text{span}\{\gamma_2, \gamma_4, \dots\}$. What is $\varphi(\mathcal{G}_n)$?

Denote

$$\mathcal{M}_n = \left\{ 1 + \sum_{i=1}^s a_i X^i : a_i \in \mathbb{F}_2 \right\} \subseteq \mathcal{R}_n.$$

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Denote

$$\mathcal{M}_n = \left\{ 1 + \sum_{i=1}^s a_i X^i : a_i \in \mathbb{F}_2 \right\} \subseteq \mathcal{R}_n.$$

Theorem

- $\mathcal{G}_n \cong \mathcal{M}_n$ as abelian monoids.
- $\mathcal{G}_n^* \cong \mathcal{M}_n^* = \mathcal{R}_n^*$ as abelian groups.

Unit groups of polynomial factor rings have an explicit description!

Fact

The unit group of the ring $\mathcal{R}_n = \mathbb{F}_2[X]/(m(X))$ is given by

$$\mathcal{R}_n^* = \{g(X) : \gcd(m(X), g(X)) = 1\}.$$

The inverse of $g(X) \in \mathcal{R}_n^$ can be found by the Extended Euclidean Algorithm.*

Theorem

Let $f = \gamma_0 + \sum_{i=1}^s \alpha_i \gamma_{2i} \in \mathcal{G}_n$.

- If n is odd, then f is a permutation on $\mathbb{F}_2^n$.
- If n is even, then f is a permutation on $\mathbb{F}_2^n$ if and only if

$$\gcd\left(1 + \sum_{i=1}^s \alpha_i X^i, X^n + X^{n/2}\right) = 1.$$

So what did we get?

$$\mathcal{G}_n = \gamma_0 + \text{span}\{\gamma_2, \gamma_4, \dots\}$$

- is a set of functions on $\mathbb{F}_2^n$
- has additive structure
- is closed under composition
- contains permutations which can be explicitly described and many of their properties can be deduced (e.g. degree, inverse, iterates, order, cycle structure)

In the previous part we had three important ingredients:

- The map χ
- The maps γ_{2i}
- The Key Lemma:

$$\gamma_{2k} \circ \left(\gamma_0 + \sum_{i=1}^s a_i \gamma_{2i} \right) = \sum_{i=0}^s a_i \gamma_{2i+2k}.$$

The first generalization appears in²:

- The map $\chi_{n,m}$ instead of χ
- Maps $\theta_{m,i}$ instead of γ_{2i}
- The identity

$$\theta_{m,k} \circ \left(\theta_{m,0} + \sum_{i=1}^s a_i \theta_{m,i} \right) = \sum_{i=0}^s a_i \theta_{m,i+k}.$$

²Cheng Lyu, Mu Yuan, Dabin Zheng, Siwei Sun, and Shun Li. “A Generalized χ_n -Function”. In: *IEEE Transactions on Information Theory* (2026). DOI: 10.1109/TIT.2026.3662335.

The map $\chi_{n,m}$

Recall:

$$\chi(x)_i = x_i + x_{i+2}(1 + x_{i+1}).$$

Definition

The map $\chi_{n,m} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n, x \mapsto y = \chi_{n,m}(x)$ for $m \geq 2$ is given by

$$y_i = x_i + x_{i+m}(1 + x_{i+m-1})(1 + x_{i+m-2}) \cdots (1 + x_{i+1}),$$

where the indices are taken modulo n .

The map $\chi_{n,m}$

Definition

$\chi_{n,m}$ is given by

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Observation

It holds that $y_i \in \{x_i, x_i + 1\}$ and

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Definition

$\chi_{n,m}$ is given by

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Observation

It holds that $y_i \in \{x_i, x_i + 1\}$ and

$$y_i = x_i + 1 \quad \Longleftrightarrow \quad (x_{i+1}, \dots, x_{i+m-1}, x_{i+m}) = (0, \dots, 0, 1).$$

Definition

$\chi_{n,m}$ is given by the complementing landscape $*0 \dots 01$.

The map $\chi_{n,m}$

Example

$n = 23, m = 3$, so we are looking for the pattern $*001$.

$$\begin{array}{ccccccccccccccccccccccc} x & = & 1 & \underline{001} & \underline{001} & 1 & 0 & 0 & \underline{001} & 0 & 1 & 0 & \underline{001} & \underline{001} & \underline{001} & 1 & 1 \\ & & \downarrow & & \downarrow & & & & \downarrow & & & & \downarrow & & \downarrow & & \\ \chi_{23,3}(x) & = & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{array}$$

The map $\chi_{n,m}$

Example

$n = 23, m = 3$, so we are looking for the pattern $*001$.

$$\begin{array}{ccccccccccccccccccccccc} x & = & 1 & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{blue}{1} & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{blue}{1} & 1 & 0 & 0 & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{blue}{1} & 0 & 1 & 0 & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{blue}{1} & \textcolor{red}{0} & \textcolor{red}{0} & \textcolor{blue}{1} & 1 \\ & & \downarrow & & & \downarrow & & & & \downarrow & & & & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & \\ \chi_{23,3}(x) & = & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{array}$$

Red positions cannot change under application of $\chi_{n,3}$, the pattern 001 itself prevents this.

Blue positions may change, depending on the rest of the vector.

Quasi-conserved landscapes

Another pattern: f given by $0 * _10$. The symbol $_$ is a placeholder meaning "this entry does not matter". As a formula:

$$f(x)_i = x_i + (1 + x_{i-1})x_{i+2}(1 + x_{i+3}).$$

Example

$$\begin{array}{cccccccccccccccccccc} x & = & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ & & & & \downarrow & & & & & & \downarrow & \downarrow & & & & & & \downarrow & & & & \\ f(x) & = & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \end{array}$$

Quasi-conserved landscapes

Another pattern: f given by $0 * \textcolor{red}{_1}0$. The $_$ means "this entry does not matter". As a formula:

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Example

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Red positions cannot change under application of f , the pattern $0 * \textcolor{red}{_1}0$ itself prevents this.

Blue positions may change, depending on the rest of the vector.

Quasi-conserved landscapes

A landscape $c = a_{-k} \dots a_{-1} * a_1 \dots a_m$ is a string consisting of 0, 1 and _.

Definition

A quasi-conserved landscape is a landscape with at most one blue position.

We can define a family of functions $\gamma_0, \gamma_1, \gamma_2, \dots$

Definition

We say that the functions γ_i satisfy the polynomial composition property if

$$\gamma_k \circ \left(\gamma_0 + \sum_{i=1}^s a_i \gamma_i \right) = \sum_{i=0}^s a_i \gamma_{i+k}.$$

Main Theorem

The following are equivalent:

- c is a quasi-conserved landscape.
- The functions γ_i satisfy the polynomial composition property.

- Pick a quasi-conserved landscape c .
- Obtain functions $\gamma_0, \gamma_1, \gamma_2, \dots$
- They automatically satisfy the Key Lemma

$$\gamma_k \circ \left(\gamma_0 + \sum_{i=1}^s a_i \gamma_i \right) = \sum_{i=0}^s a_i \gamma_{i+k}.$$

- Thus: The set $\mathcal{G}_n^{(c)} = \gamma_0 + \text{span}\{\gamma_1, \gamma_2, \dots\}$ is a monoid and we can use the same ideas as before.
- We obtain large families of shift-invariant permutations on $\mathbb{F}_2^n$.

For more details about χ :

- n odd: Björn Kriepke and Gohar Kyureghyan. “Algebraic Structure of the Iterates of χ ”. In: *Advances in Cryptology – CRYPTO 2024*. Ed. by Leonid Reyzin and Douglas Stebila. Cham: Springer Nature Switzerland, 2024, pp. 412–424. ISBN: 978-3-031-68385-5
- n even: Björn Kriepke and Gohar Kyureghyan. “There are siblings of χ which are permutations for n even”. In: *Des. Codes Cryptogr.* 94.6 (June 2026). DOI: 10.1007/s10623-026-01830-0

About quasi-conserved landscapes: In preparation

Thank you for your attention.